

KILLEEN REGIONAL, TX, USA

WMO: 722576

Lat: 31.067N

Lon: 97.833W

Elev: 309

StdP: 97.66

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 03902

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.7	-1.5	-11.8	1.4	2.1	-9.3	1.8	4.5	12.0	10.8	11.0	12.3	4.7	0	0.490

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.0	38.0	22.8	37.2	22.9	36.0	22.9	25.4	32.1	24.9	31.5	24.5	30.9	4.5	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.9	19.5	27.1	23.3	18.7	26.7	22.9	18.3	26.5	79.4	31.8	77.2	31.3	75.4	30.7	30.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.7	9.3	8.4	-6.7	40.3	2.3	1.8	-8.4	41.6	-9.7	42.7	-11.0	43.7	-12.7	45.0		
(5)			-7.9	26.7	2.2	1.1	-9.4	27.5	-10.7	28.2	-11.9	28.8	-13.5	29.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	20.0	9.9	12.0	15.6	19.9	24.0	27.3	29.4	29.6	26.2	20.8	14.8	10.5
	DBStd	8.23	5.74	6.00	5.13	4.11	3.33	2.22	2.03	2.08	3.28	4.36	5.20	5.36
	HDD10.0	211	73	43	15	1	0	0	0	0	0	1	17	59
	HDD18.3	1003	266	188	112	30	3	0	0	0	1	27	126	249
	CDD10.0	3876	71	99	188	299	433	521	601	607	484	337	162	74
	CDD18.3	1627	4	11	27	78	178	271	343	348	236	104	21	6
	CDH23.3	17911	31	83	199	617	1619	2974	4376	4493	2497	863	131	29
CDH26.7	8829	3	22	40	183	644	1446	2436	2535	1204	297	18	2	
Wind	WSAvg	4.3	4.5	4.6	4.9	5.1	4.7	4.4	4.0	3.7	3.4	3.9	4.2	4.2
Precipitation	PrecAvg	825	50	57	73	65	110	83	50	53	72	91	69	53
	PrecMax	1227	123	208	253	156	261	219	184	179	189	270	277	209
	PrecMin	424	0	0	0	10	29	8	0	4	8	5	5	3
	PrecStd	205	35	47	51	40	59	56	45	49	48	69	59	49
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.6	29.3	29.9	33.7	36.2	38.0	39.9	39.9	38.1	33.5	28.9	26.5
		MCWB	15.6	15.7	18.1	20.6	22.2	23.4	22.5	22.6	22.4	21.0	19.7	17.2
	2%	DB	23.6	25.8	27.6	30.9	33.8	36.0	37.9	38.1	36.1	31.6	26.7	23.7
		MCWB	15.5	15.5	17.8	20.2	22.6	23.7	22.8	22.7	22.6	20.6	18.7	16.7
	5%	DB	21.2	23.0	25.7	28.7	32.2	34.4	37.1	37.3	34.2	30.0	24.9	21.3
		MCWB	14.8	15.6	17.4	19.8	22.3	23.2	23.0	22.8	22.2	20.3	18.2	15.7
	10%	DB	19.0	21.0	23.5	27.0	30.2	32.9	35.7	36.0	32.6	28.0	22.8	19.1
		MCWB	14.2	15.5	16.7	19.0	21.9	23.2	23.1	22.9	22.1	19.6	17.4	14.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.9	20.1	21.1	23.9	25.8	26.0	25.7	25.7	25.7	24.5	22.4	20.6
		MCDB	22.0	22.7	25.8	28.8	31.4	32.9	33.0	32.8	31.9	28.4	26.0	22.7
	2%	WB	18.4	19.1	20.1	22.8	24.5	25.2	25.1	25.0	24.9	23.5	21.1	19.1
		MCDB	21.1	21.7	23.9	27.6	29.7	31.9	32.5	32.0	30.4	27.4	24.1	21.8
	5%	WB	17.0	18.1	19.4	21.8	23.8	24.6	24.7	24.5	24.2	22.7	20.0	17.4
		MCDB	19.7	20.8	22.8	25.9	28.8	31.0	31.8	31.5	29.6	26.7	22.7	20.0
	10%	WB	14.9	16.8	18.6	21.0	23.0	24.1	24.2	24.1	23.6	21.9	18.9	15.6
		MCDB	17.9	20.0	21.7	24.6	28.0	30.0	31.1	31.0	29.0	25.8	21.9	18.2
Mean Daily Temperature Range	5% DB	MDBR	11.5	11.5	11.6	11.6	10.9	10.8	11.7	12.0	11.4	11.3	11.0	11.0
		MCDBR	14.3	13.8	13.6	13.4	12.8	12.4	13.6	13.6	13.0	12.8	12.4	13.3
		MCWBR	8.3	7.8	7.0	6.5	4.9	3.4	3.4	3.3	4.0	5.6	6.7	8.1
	5% WB	MCDBR	11.7	10.9	10.7	10.8	10.7	10.5	11.5	11.5	10.4	10.0	10.0	11.1
		MCWBR	8.5	8.0	6.7	6.3	5.2	3.7	3.4	3.3	3.8	5.5	7.2	8.5
Clear-Sky Solar Irradiance	taub		0.312	0.325	0.351	0.391	0.425	0.420	0.432	0.423	0.394	0.366	0.340	0.315
	taud		2.521	2.488	2.428	2.338	2.297	2.352	2.355	2.363	2.393	2.457	2.471	2.515
	Ebn at Noon		921	938	932	901	868	867	855	861	876	880	878	897
	Edh at Noon		86	98	111	127	133	125	125	122	114	100	90	83
All-Sky Solar Radiation	RadAvg		3.03	3.55	4.58	5.51	5.83	6.66	6.74	6.28	5.22	4.21	3.19	2.68
	RadStd		0.32	0.58	0.41	0.50	0.54	0.51	0.52	0.36	0.46	0.62	0.39	0.36

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-207	-166
Regional (44 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+73

Nomenclature: See separate page